

MJ Team • CSIS 4495

3D Forge

Type an idea, get a **3D model** — then **texture**, **segment**, and **edit it part-by-part**, all in the browser.

DEVELOPER

Mikhail Senatorov

DEVELOPER

Javid Aliyev

Gemini · image

Tripo · text→3D · segment

Three.js · viewer

React + Express

MongoDB

HERO IMAGE

ZiWatXqbKwdLVWcDwlb83M2iKxm1VYTm/Pi+CoCJtOfkPK2apEW+rKp41oPypapcPe
(segmented rainbow robot)

HOW IT WORKS

One **pipeline**: prompt → image → 3D → edit

The user always sees — and **approves the picture** — before spending a 3D generation. Verify on the image, then commit to the model.



Honest note: a rebuilt part is a new mesh, so re-inserting it can need a little alignment — an open research nuance we show live.

► Publish to the community gallery

From raw mesh to editable parts

1 · Generate

Vgc3t3efefLMtLDXC8v3/SRJVNEMZTwJjk+Ct0n7+XhTIKlnaZZtbGxEUel5oLDKM5TKzFLYfyrRzjBFS7cN
(grey untextured robot)

oR9R0DSliO20W9DX+bljGdosDMmwfXGKkkWB
(red robot)

3kB34UhIYFaR9tQhjnSg0GXALfd3f7A4PSWqN57F
(white/blue robot)

Generate — prompt → image → raw 3D mesh

Recolor — glossy red

Recolor — clean white

2 · Segment

gh8blVolZfm9iXaA/2jCtf5ZGWO+JGY+OouazWV
(parts list panel)

63Pvn90tL09ur49uj4OtuL40rA/SZLdo8m9IDiS8u
(exploded parts)

zQZKtTW5KnDogk+kpGqheBxbcQTXBWYPOSgB
(part isolated)

κ0ozTjIbRJGNLQ0IEzSYeprSUibtC8RUSBgoFNZ7F
(part re-edited)

Segment — Tripo splits into parts

Explode — inspect each part

Isolate — one segment

Rebuild — edited & stitched